University of Rajshahi
Department of Computer Science and Engineering
B. Sc. (Engg) Part-IV Odd Semester Examination 2024
Course: CSE-4111 (Parallel Processing and Distributed System)
Session:2020-2021, 2019-2020 and 2018-2019
Full Marks: 52.5 Duration: 3(Three) Hours

Answer 06(Six) questions taking any 03(Three) questions from each section

Section – A

1. a) Define Parallel System and Distributed System. What are their key differences? 4.00
   b) Why does increasing the number of processors not always improve performance? 2.75
   c) Write some applications of parallel processing. 2.00

2. a) Draw and discuss the basic structure of an Array Computer? 2.00
   b) Discuss design space of major **processor** families. 2.75
   c) Briefly explain the architecture of a superscalar processor. 4.00

3. a) Classify pipeline processors according to the levels of processing. 1.75
   b) Draw and explain the structure of a pipeline floating point adder. 3.50
   c) What is Amdahl's Law, and how does it limit the theoretical speedup of a parallel program? Illustrate with a numerical example. 3.50

4. a) Define pipenet. Show the program graph and the pipenet of following equation 3.00
   $$E(I) = [A(I)\times B(I) + B(I)\times C(I)]/[B(I)\times C(I)\times(C(I) + D(I))]$$
   b) Draw the interconnection structure of a generalized multiprocessor system. 2.00
   c) What is program graph transformation? Write the lemmas to convert a given program graph into an equivalent graph. 2.75
   d) What is meant by resource conflict? 1.00

Section – B

5. a) What are the major principles for designing a vectorizer? 2.00
   b) State different vector instruction types with example. 3.00
   c) Explain the optimization of vector functions with example. 3.75

6. a) Distinguish between local and remote distribution. 2.00
   b) Write the main factors that trend to distributed processing. 3.00
   c) Briefly explain hierarchical distributed processing with example. 3.75

7. a) What is fragmentation? Differentiate between horizontal and vertical fragmentation with example. Also write their advantages. 3.00
   b) What is replication? Write about some advantages and disadvantages of data replication. 2.25
   c) Why do we use commit protocol? Briefly explain two-phase commit protocol. 3.50

8. a) Write the role of the transaction manager and transaction coordinator for a distributed system. How do they ensure consistency across multiple systems? 3.75
   b) What are the main reasons for failure of a distributed transaction? 2.50
   c) How does Heterogeneous Distributed Databases differ from Homogeneous Distributed Databases? 2.50

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-4 Odd Semester Examination-2023
Course: CSE4111 (Parallel Processing and Distributed System)
Marks: 52.50 Time: 3:00 Hours

[N.B. Answer any Six questions taking Three from each section]

Section-A

1.	(a)	Define the terms: (i) Parallel System (ii) Distributed System (iii) Multiprogramming (iv) Timesharing (v) Multi-processing.	3.75
	(b)	What are the mechanisms of implementing parallel processing in uni-processor system?	3
	(c)	Distinguish between parallel and distributed data processing.	2
2.	(a)	How does parallel computing modify the traditional von Neumann model?	2
	(b)	Compare and contrast different types of parallel hardware architectures (e.g., SMP, MPP, GPU-based processing). Provide examples of their applications.	4
	(c)	What challenges arise in parallel program design, and how can they be addressed?	2.75
3.	(a)	Define linear, non-linear, synchronous, and asynchronous pipeline processor with necessary diagram.	4
	(b)	What is interleaved memory? Why is interleaved memory beneficial for vector processors and parallel computing?	2.75
	(c)	A system has 4-way interleaved memory, with each memory module having an access time of 50 ns and a cycle time of 200 ns. (i) Calculate the effective memory bandwidth if the processor issues memory requests every 50 ns. (ii) If the system increases to 8-way interleaving, what is the new effective bandwidth?	2
4.	(a)	Explain the working principle of a Carry Save Adder with an example of three 4-bit numbers.	1.5
	(b)	Define horizontal processing, vertical processing and vector looping in terms of pipelined vector processing method.	2.25
	(c)	Compute the speed-up of each processing methods with respect to scalar processor for the following linear combination of n vector $y=a_1.x_1+a_2.x_2+a_3.x_3+\ldots+a_n.x_n$; where $\{a_i$ for $1 \le i \le n\}$ be n scalar constant, $y = (y_1, y_2, \ldots, y_m)^T$ be a column vector for m components, and $x_j = (x_{1j}, x_{2j}, \ldots, x_{mj})^T$ for $j = 1, 2, \ldots, n$ be n column vector. Given that, setup time and required time for addition/multiplication equals $5t$ where t is the clock period.	5

Section-B

5.	(a)	Define vector and stride.	2
	(b)	What do you mean by vector task system? Explain briefly with example.	5
	(c)	How could we enhance vector-processing capability?	1.75
6.	(a)	Define software path length and teleprocessing.	1
	(b)	State and explain different categories of data distribution.	4
	(c)	What are the main reasons of function distribution?	3.75
7.	(a)	What is a Distributed Database and how does it differ from a centralized database?	1.5
	(b)	Distinguish between homogeneous and heterogeneous distributed database.	2.25
	(c)	Define data fragmentation in a Distributed Database. Explain various types of fragmentation with example? Also write their advantages.	5
8.	(a)	Define the three conditions of parallelism: data dependence, control dependence, and resource dependence.	2
	(b)	What are the differences between static scheduling and dynamic scheduling?	1.75
	(c)	Describe various interconnect architectures (Bus, Ring, Mesh, Hypercube, Crossbar) and their advantages/disadvantages.	5

	University of Rajsahhi	
	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING	
	B.Sc. Engineering Part IV Odd Semester Exam-2022	
	Course Code: CSE4111 Course Title: Parallel Processing and Distributed System	
Time: 03 Hours		Full Marks: 52.50

SECTION A
ANSWER ANY THREE OF THE FOLLOWINGS

1.(a) Why are we developing parallel systems instead of sequential systems? [3.00]
(b) Why do we need to write parallel programs based on parallel algorithms? [3.25]
(c) How can we exploit instruction level parallelism and loop level parallelism? [2.50]

2.(a) How Flynn classifies computers? Discuss with necessary diagrams. [3.00]
(b) Differentiate between the characteristics of CISC and RISC processor architectures. [2.00]
(c) Briefly explain the architecture, instruction format, and pipeline operations of a typical VLIW processor with necessary diagram. [3.75]

3.(a) Define speedup, efficiency and throughput. [2.25]
(b) Define interleaved memory organization. Compare between high-order memory interleaving and low-order memory interleaving? [3.50]
(c) Draw the interconnection structure of a generalized multiprocessor system. [3.00]

4.(a) Classify pipeline processors according to the levels of processing. [1.75]
(b) Draw and explain the structure of a pipeline floating point adder. [3.50]
(c) Explain the functional segmentation of typical storage-to-register floating point instruction in the IBM 360/91. [3.50]

SECTION B
ANSWER ANY THREE OF THE FOLLOWINGS

5.(a) What is false sharing? Explain with an example. [2.75]
(b) What are the differences among mutex, busy-waiting and semaphore? [3.00]
(c) What is the purpose of using Allgather() function in MPI? [3.00]

6.(a) Distinguish between local and remote distribution. [1.50]
(b) Write the main factors that trend to distributed processing. [3.00]
(c) What are the basic differences between function and system distribution? [2.00]
(d) Write different categories of data distribution. [2.25]

7.(a) Write the role of transaction manager and transaction coordinator for a distributed database. [2.75]
(b) What is fragmentation? Differentiate between horizontal and vertical fragmentation with example. Also write their advantages. [3.50]
(c) What are the main reasons of failure of a distributed transaction? [2.50]

8.(a) Briefly explain two-phase commit protocol. [4.00]
(b) What is replication? Write some advantages and disadvantages of data replication. [2.25]
(c) Define Heterogeneous Distributed Databases. Write its advantages [2.50]

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

University of Rajshahi

B.Sc. Engineering Part IV Odd Semester Examination-2021

Course Code: CSE4111 Course Title: Parallel Processing and Distributed System

Time: 03 Hours Full Marks: 52.50

SECTION A
[ANSWER ANY THREE OF THE FOLLOWINGS]

1.(a) *'Parallel computing is the simultaneous use of multiple computing resources to solve a computational problem'*. Give some real-life examples. Write the main challenges for parallel computing. [3.50]

(b) What are the levels of parallel processing? Explain each of the levels with at least one example. [4.00]

(c) Does parallel processing differ with multiprocessing? Explain your answer. [1.25]

2.(a) What do you mean by **'Flynn's** Taxonomy'? Explain how SIMD and MIMD computers work with instructions and data. [4.50]

(b) *'Parallelism can be achieved in uniprocessor system using some hardware and software technologies'*- what are they? [2.00]

(c) Explain how parallelism can be achieved using ***multiprogramming*** and ***time-sharing*** methods. [2.25]

3.(a) *'Data dependency is a situation in which a program statements refer to the data of a preceding statement'*. What are the types of it? Explain with examples. [4.00]

(b) Consider the following code fragment of a program— [2.75]

S1: Load R1,A
S2: Add R2,R1
S3: Move R1,R3
S4: Store B,R1

Find the dependencies between the statements in the code fragment.

(c) Explain how ***I2*** is resource dependent on ***I1*** in the fragment given below— [2.00]

I1: A=B+C
I2: D=G+H

Hence, draw the dependency graph.

4.(a) Define the following terms: (i) Speedup, (ii) Efficiency, (iii) Throughput. [3.00]

(b) A five-stage linear pipeline has clock period of τ=100ns, and n=50 tasks. What is its throughput, efficiency and speedup? [3.25]

(c) A CPU of modern computer is a good example of linear pipeline. Explain. [2.50]

SECTION B
[ANSWER ANY THREE OF THE FOLLOWINGS]

5.(a) What are the abstract parallel programming models over the hardware and memory architectures? Compare the relative advantages and disadvantages of ***shared memory model*** with ***message passing model***. [4.75]

(b) What do you mean by MPI? Discuss the purposes and uses of the followings— [4.00]
i) MPI_Init, ii) MPI_Finalize, iii) MPI_Send and iv) MPI_Recv

6.(a) What is cache coherency? With a suitable example, explain how cache becomes incoherent due to the ***write-access*** of shared variables from multiple number of processors. [4.00]

(b) When two processors ***P1*** and ***P2*** have same data element ***X*** in their local caches and one processor modifies the value of ***X*** to ***Y***, then the processor's cache becomes inconsistent with the other's processor's cache and the shared memory. Explain this inconsistency problem for the situations of both ***data Sharing*** and ***process Migration,*** in terms of ***write-through*** and ***write-back*** policies. Use suitable examples and figures. [4.75]

7.(a) Suppose you have an array {12, 14, 6, 8, 10, 16, 2, 4}, show all the phases of sorting the array using odd-even transposition sort algorithm. [4.50]

(b) Write a program using OpenMP to implement the odd-even transposition sort algorithm. [4.25]

8.(a) Define linear and non-linear pipelines. Discuss their differences. [3.75]

(b) Consider the following reservation table of a three-stage non-linear pipeline to execute a function **X**. [5.00]

	Time---->								
		1	2	3	4	5	6	7	8
Stage	S1	X					X		X
	S2		X		X				
	S3			X		X		X	
	Reservation Table for function **X**								

i) What are the forbidden and permissible latencies?

ii) Calculate the collision vector.

iii) Draw the state transition diagram.

University of Rajshahi

Department of Computer Science & Engineering

B.Sc. Engineering Part IV Odd Semester Examination 2020

Course Code: CSE 4111

Course Title: Parallel Processing and Distributed System

Full Marks: 52.5 **Time: 3 Hours**

[Answer any **SIX (06)** questions taking **THREE (03)** from each section]

Section A

1.	a)	Why are we developing parallel systems instead of sequential systems?	3
	b)	Why do we need to write parallel programs?	2.75
	c)	How can we exploit instruction level parallelism and loop level parallelism?	3
2.	a)	If we fetch a cache line from main memory, where in the cache should it be placed?	3.25
	b)	What is the bottleneck of Von Neumann architecture?	1
	c)	Explain which function is **thread safe** and which is not.	2
	d)	What are the limitations of MPI_Scatter and MPI_Gather functions?	2.50
3.	a)	What are the key characteristics of a vector processor?	2.75
	b)	Does false sharing generate incorrect result? Explain the effects of false sharing.	3
	c)	What is cache coherence problem? Write a program to illustrate the cache coherence problem.	3
4.	a)	How can the receiver find out the following values in MPI? 1. The amount of data in the message. 2. The sender of the message. 3. The tag of the message.	3.75
	b)	Why do we need MPI_Reduce and MPI_Allreduce function? Show the communication pattern of MPI_Allreduce.	3
	c)	Write short notes on GPU.	2

Section B

5.	a)	Why do we use MPI_Ssend instead of MPI_Send?	2
	b)	A set of numbers (15, 11, 9, 16, 3, 14, 8, 7, 4, 6, 12, 10, 5, 2, 13, 1) and four processes are given. Show how the processes can sort the numbers using parallel odd-even transposition sort.	3.25
	c)	Write a program in MPI to sort a set of numbers using parallel odd-even transposition sort algorithm.	3.50
6.	a)	Define grain size and latency.	2
	b)	Define different types of data dependence.	3.25
	c)	What are the steps of grain packing? Explain with an example.	3.50
7.	a)	Can multiple threads simultaneously insert new nodes into a linked list without any problem? If not, explain how to insert the nodes and write a code segment to insert the nodes by multiple threads.	3.25
	b)	How can we synchronize threads?	3
	c)	What is mutex? Write a program using mutex.	2.50

8. a) What are the problems of busy-waiting? 2.50

b) Define critical section and race condition. 2.50

c) fibo[0] = fibo[1] = 1; 3.75

```
# pragma omp parallel for num_threads(thread_count)
for (i = 2; i < n; i++)
fibo[i] = fibo[i-1] + fibo[i-2];
```

What will be the results if we try running the above code with more than one thread?

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. (Engg.) Part-4, Odd Semester, Examination-2019

Course: CSE-4111 (Parallel processing and distributed system)

Time: 03 Hours Full Marks: 52.5

Section A

Answer any THREE questions.

1. (a) What do you mean by parallel processing? 1
   (b) Discuss different types of data dependence. 2
   (c) Two 3x3 matrices A and B are given to compute the sum of the nine elements in the resulting product matrix C=AxB. On the other hand, 150 cycles for inter-processor communication latency, 80 cycles for the nodes performing multiplication, 15 cycles for the nodes performing addition are given. Draw a fine-grain program graph. Perform grain packing and parallel scheduling. 3.5
   (d) What is data-flow computer? Write a simple C program and draw its dataflow graph. 2.25

2. (a) How can you define speedup factor, efficiency and throughput of a k-stage pipeline processor over an equivalent non pipelined processor? 2.75
   (b) 5

	1	2	3	4	5	6
S1	X					X
S2		X			X	
S3			X			
S4				X		
S5		X				X

Considering the above reservation table

(a) List the set of forbidden latencies and the collision vector

(b) Draw the state transition diagram

(c) Determine the minimum average latency of this pipeline

   (c) Write some applications of vector processor? 1

3. (a) Define instruction set. Briefly discuss complex instruction set and reduced instruction set. 4.75
   (b) Differentiate among scalar, superscalar, and vector processor. 2
   (c) Which type of memory interleaving can tolerate memory fault and which cannot? Explain. 2

4. (a) What is CUDA kernel function? 1.5
   (b) What are the effects of launching CUDA kernel functions? 2.25
   (c) Two vectors are given. Write a CUDA kernel to perform addition of two vectors. 3
   (d) How can you set grid and thread block dimensions in CUDA programming? 2

Section B

Answer any THREE questions.

5. (a) Define bisection width and bisection bandwidth. 2
   (b) Discuss snooping cache coherence and directory-based cache coherence method. 4
   (c) Write down the Amdahl's law. 2.75

6. (a) Write a program to read a vector and distribute it among processes using MPI provided function. 3.25
   (b) Sort the numbers 15, 11, 9, 16, 3, 14, 8, 7, 4, 6, 12, 10, 5, 2, 13, 1 using parallel odd-even transposition sort algorithm. 3.5
   (c) What is the purpose of using Allgather() function in MPI? 2

7. (a) What is false sharing? Explain with an example. 2.25
   (b) What are the differences among mutex, busy-waiting and semaphore? 3
   (c) Can multiple threads simultaneously delete nodes of a linked list without any problem? If not, explain how to delete the nodes and write a code segment to delete the nodes by multiple threads. 3.5

8. (a) Mention the desired features of a parallel language. 2
   (b) Convert the following sequential code to vector code: 2

```
DO 20 I=1, N
        A(I)=B(I-1)
20 20 B(I)=2*B(I)
```

   (c) Describe the following optimization of vector functions: 2.75+2
      (i) Redundant express elimination.
      (ii) Constant folding at compile time